

MATSU NANGAN, Taiwan

WMO: 466890

Lat: 26.160N

Lon: 119.958E

Elev: 68

StdP: 100.51

Time Zone: 8.00 (E08)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	5.6	-0.3	3.7	7.4	0.9	4.1	7.4	15.5	6.5	14.3	6.9	8.6	30	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	31.6	27.7	30.9	27.5	30.0	27.1	28.2	30.5	27.7	29.9	27.2	29.3	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	23.8	30.3	27.1	23.0	29.8	26.6	22.3	29.1	91.9	30.9	89.3	30.3	87.1	29.5	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.6	12.0	2.7	33.1	1.7	1.5	1.5	34.2	0.5	35.0	-0.4	35.9	-1.6	37.0		
(5)			1.3	29.5	0.9	0.5	0.6	29.9	0.1	30.1	-0.4	30.4	-1.1	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	10.0	9.9	11.7	15.8	20.8	24.8	27.4	27.5	25.6	21.5	17.7	13.3	(6)
(7)		DBStd	6.82	2.71	2.64	2.64	3.21	2.64	2.48	1.25	1.18	1.85	2.25	2.65	2.46	(7)
(8)		HDD10.0	82	34	31	12	1	0	0	0	0	0	0	0	4	(8)
(9)		HDD18.3	997	257	236	207	88	8	0	0	0	0	3	42	157	(9)
(10)		CDD10.0	3322	35	28	64	175	334	443	541	542	467	356	231	105	(10)
(11)		CDD18.3	1197	0	0	1	12	84	193	282	283	217	101	23	1	(11)
(12)		CDH23.3	9287	0	0	1	12	236	1386	2884	2933	1633	193	8	1	(12)
(13)		CDH26.7	2209	0	0	0	1	24	291	795	800	291	7	1	0	(13)
(14)	Wind	WSAvg	5.9	7.3	6.8	6.2	4.9	4.7	4.6	4.3	4.7	5.6	7.3	7.6	7.3	(14)
(15)	Precipitation	PrecAvg	1731	71	96	145	151	213	260	178	259	166	66	69	58	(15)
(16)		PrecMax	2408	218	254	317	341	392	445	430	498	574	191	226	193	(16)
(17)		PrecMin	938	9	19	53	52	46	80	21	66	25	2	5	5	(17)
(18)		PrecStd	335	50	57	61	61	84	83	101	130	125	56	51	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	18.9	21.0	24.8	29.0	31.9	32.3	32.2	31.8	27.3	24.2	21.0	(19)
(20)		MCWB	15.6	16.4	18.7	22.4	25.5	27.9	28.1	27.8	27.8	23.5	21.1	18.3	(20)	
(21)		2%	DB	16.2	16.8	18.9	23.1	27.0	30.2	31.7	31.3	30.2	26.2	22.9	18.9	(21)
(22)		MCWB	13.4	15.1	17.0	21.1	24.0	27.2	27.8	27.6	27.2	22.9	20.5	16.9	(22)	
(23)		5%	DB	15.0	15.2	17.2	21.8	25.6	29.2	30.9	30.8	29.1	25.2	22.0	17.8	(23)
(24)		MCWB	12.7	13.7	15.8	20.0	23.2	26.7	27.5	27.5	26.3	22.5	19.9	16.0	(24)	
(25)		10%	DB	13.9	13.8	15.9	20.5	24.5	28.2	30.0	30.0	28.2	24.2	21.1	16.8	(25)
(26)		MCWB	11.9	12.2	14.5	19.0	22.5	26.0	27.1	27.1	25.6	21.7	19.1	15.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	17.1	18.8	22.7	25.9	28.5	29.2	29.0	28.5	25.0	22.1	18.6	(27)
(28)		MADB	18.2	18.4	20.6	24.1	28.3	30.9	31.7	31.2	30.7	26.4	23.0	20.2	(28)	
(29)		2%	WB	14.0	15.3	17.4	21.4	24.6	27.5	28.2	28.2	27.5	24.1	21.1	17.3	(29)
(30)		MADB	15.9	16.6	18.6	22.9	26.4	29.7	30.5	30.5	29.6	25.5	22.3	18.4	(30)	
(31)		5%	WB	13.0	13.9	16.1	20.3	23.7	26.7	27.7	27.6	26.7	23.1	20.3	16.2	(31)
(32)		MADB	14.7	15.3	17.1	21.6	25.2	28.7	29.9	29.9	28.4	24.7	21.6	17.3	(32)	
(33)		10%	WB	12.0	12.5	14.7	19.1	22.9	26.1	27.1	27.1	26.1	22.1	19.4	15.1	(33)
(34)		MADB	13.7	13.5	15.7	20.4	24.2	27.8	29.3	29.3	27.7	24.0	20.9	16.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	4.4	4.4	4.8	5.3	5.3	5.6	5.4	5.1	4.5	4.0	4.1	4.1	(35)	
(36)		5% DB	MCDBR	6.8	7.9	7.8	7.3	6.9	6.5	6.7	6.3	5.8	5.0	5.3	6.1	(36)
(37)		MCWBR	5.2	6.2	6.2	5.7	5.2	4.8	4.3	4.1	3.8	3.6	3.8	4.9	(37)	
(38)		5% WB	MCDBR	6.2	7.2	7.1	7.3	6.6	6.2	5.8	5.7	5.4	4.6	4.8	5.9	(38)
(39)		MCWBR	5.1	6.2	6.3	6.0	5.4	4.7	4.1	4.0	4.0	3.8	4.1	5.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.580	0.633	0.708	0.707	0.623	0.572	0.541	0.571	0.564	0.571	0.562	0.555	(40)	
(41)		taud	1.758	1.685	1.572	1.628	1.846	1.998	2.088	1.993	1.983	1.888	1.867	1.831	(41)	
(42)		Ebn at Noon	653	657	637	656	714	747	770	745	737	700	666	653	(42)	
(43)		Edh at Noon	199	230	270	262	210	179	164	179	177	185	177	178	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.21	2.54	3.28	3.73	4.19	4.61	6.07	5.45	4.51	3.65	2.51	2.27	(44)	
(45)		RadStd	0.52	0.54	0.38	0.49	0.68	0.51	0.48	0.46	0.44	0.60	0.39	0.42	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	+77

Nomenclature: See separate page